Robert Tylman

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EXPERIENCE

Nebius (Tavily) - Forward Deployed Engineer *Manhattan, NY · Aug 2026 – Present*

- Designing and deploying production integrations of Tavily's web search API for commercial and large enterprise clients.
- Creating POCs, product demos, and conducting benchmarks to quantify Tavily's performance and accelerate wins.
- Devising several new agentic endpoints for search/research, owning the process from prototype to production.

Veritonic - Machine Learning Intern *Manhattan, NY · May 2026 – Jul 2026*

- Shipped a new product for real-time audio emotion classification and sentiment analysis using CLAP embeddings with added fine-tuning.
- Developed and deployed numerous key features for existing products in Veritonic's audio intelligence platform.

Camp Solelim — Assistant Director & Head Tripper *Sudbury, Canada · May 2021 – Aug 2025*

- Developed an internal logistics and route-planning platform leveraging Google Earth APIs for campsite optimization, meal planning, and route visualization, reducing seasonal costs by $10K+ and increasing trip enrollment by 500%.
- Created *CampSorter*, a full-stack scheduling and logistics platform, reducing admin workload by 75% and improving camp programming efficiency; system remains in active use.

People's Trust Company — Software Engineer & IT Analyst Intern *Toronto, Canada · May 2023 – Aug 2023*

- Partnered with finance and operations teams to audit their manual data workflows, then built Python automation pipelines that replaced the most error-prone steps and cut input errors by 95%.
- Owned company-wide IT infrastructure for 600+ employees: encryption, networking, onboarding, permissions, security.

MyPart — Machine Learning Intern *Tel Aviv, Israel · Sep 2020 – May 2021*

- Refined retrieval and feature-extraction pipelines over a 250,000+ track dataset, then trained multimodal embedding models for recommendation and semantic similarity ranking, improving retrieval relevance by 15% over baseline.

PROJECTS

Ableton MCP Server — *Ableton Live API, Agents, MCP, Python, Socket/UDP*

- Collaborated on a production MCP server exposing Ableton Live as a tool surface for agents (Claude, Codex); designed the tool schema and state layer so natural-language prompts translate into real-time DAW operations (procedural audio generation, track organization, track automation, MIDI programming, device manipulation).

Twist! — *Expo, Gemini 3.1 API, Node.js, React, Redis, Socket.io*

- Shipped a production multiplayer party-game app on a 4-developer team where users vote on AI-generated image transformations of their own photos; reached 350+ players and $100+ revenue in the first month.
- Designed disconnection recovery using heartbeat-based liveness detection and cross-state server synchronization.

Multimodal Audio-to-Text Captioning Model — *BART, BEATs, CLAP, ConvNeXt, PyTorch, Transformers*

- Built a multimodal encoder–decoder system that generates highly descriptive audio captions by aligning audio and text in a shared embedding space for semantic understanding.
- Implemented a multi-stage, multi-encoder pipeline with diverse candidate generation (nucleus sampling), contrastive reranking with CLAP, and LLM-based refinement for final caption synthesis.

EDUCATION

New York University — New York, NY *2025 – 2026*

M.S., Machine Learning — Music Information Retrieval. GPA 4.00; ***Sony Audio Institute Scholarship***.
Research: Multimodal representation learning with pretrained encoders for audio understanding.

Vanderbilt University — Nashville, TN

B.S., Computer Science (Engineering) — GPA 3.83 (Cum Laude, Dean's List).
2023 Blair Award for Innovative Work in Audio Technology.

SKILLS

AI & Agent Systems: Embedding/Vector Databases, Hybrid Retrieval, LangChain, LLM Agents, MCP, Ragas, RAG Pipelines
Engineering: AWS, C++, Docker, Expo, Git, HPC, JavaScript, Librosa, Node.js, Python, React, Redis, Socket.io, SQL, TypeScript